Isoperimetric Inequalities and the Width Parameters of Graphs*

L. Sunil Chandran T. Kavitha**
C. R. Subramanian***

Abstract. We relate the isoperimetric inequalities with many width parameters of graphs: treewidth, pathwidth and the carving width. Using these relations, we deduce

1. A lower bound for the treewidth in terms of girth and average degree
2. The exact values of the pathwidth and carving width of the d–dimensional hypercube, H_d
3. That treewidth $(H_d) = \Theta\left(\frac{2^d}{\sqrt{d}}\right)$.

Moreover we study these parameters in the case of a generalization of hypercubes, namely the Hamming graphs.

1 Introduction

Discrete isoperimetric inequalities have recently become important in combinatorics. Usually two kinds of isoperimetric problems are considered: the vertex isoperimetric problem and the edge isoperimetric problem. Let $G = (V, E)$ be a graph. For $S \subseteq V$, the vertex boundary $\Phi(S)$ can be defined as follows.

$$\Phi(S) = \{w \in V - S : \exists v \in S \text{ such that } \{w, v\} \in E\} \tag{1}$$

The vertex isoperimetric problem is to minimize $|\Phi(S)|$ over all subsets S of V with $|S| = \ell$ for a given integer ℓ. We denote this minimum value by $b_v(\ell, G)$, i.e., $b_v(l, G) = \min_{S \subseteq V(G), |S|=\ell} |\Phi(S)|$. (We denote by $V(G)$ and $E(G)$ the vertex and edge sets of G, respectively.)

The edge isoperimetric problem can be defined in a similar way. Given a subset S of V, its edge boundary $\delta(S)$ can be defined as

$$\delta(S) = \{e \in E : \text{exactly one end point of } e \text{ is in } S\} \tag{2}$$

* This is a combined announcement of the results in two different papers. The results explained in Section 3 is from *Girth and Treewidth* (*L. S. Chandran, C. R. Subramanian*) [8]. The rest of the results are from *Lower bounds for width parameters of graphs using Isoperimetric Inequalities* (*L. S. Chandran, T. Kavitha*) [7].

** Max–Planck–Institute für Informatik, Stuhlsatzenhausweg 85, 66123, Saarbrücken, Germany. email: {sunil,kavitha}@mpi-sb.mpg.de

*** The Institute of Mathematical Sciences, Chennai, 600113, India. Email : crs@imsc.res.in

The edge isoperimetric problem is to minimize $|\delta(S)|$ over all subsets S of V with $|S| = \ell$ for a given integer ℓ. Let $b_e(\ell, G) = \min_{S \subseteq V(G), |S|=\ell} |\delta(S)|$.

Many methods have been discovered to attack the isoperimetric inequalities – including martingale techniques [22], eigen value analysis [1] and purely combinatorial methods. (See [19] for a survey.) The discrete isoperimetric inequalities are of interest, not only because they answer extremely natural and basic questions about graphs, but also because they have numerous applications, most notably to random graphs and geometric functional analysis. In this paper, we explore yet another application of isoperimetric inequalities, namely to provide lower bounds for certain "width" parameters of graphs.

The notions of treewidth and pathwidth were introduced by Robertson and Seymour [20, 21] in their series of fundamental papers on graph minors.

Definition 1. *A tree decomposition of $G = (V, E)$ is defined to be a pair (X, T) where $X = \{X_i : i \in I\}$ is a collection of subsets of V (we call these subsets the nodes of the decomposition) and $T = (I, F)$ is a tree having the index set I as the set of vertices such that the following conditions are satisfied:*

1. $\bigcup_{i \in I} X_i = V$.
2. $\forall (u, v) \in E, \exists i \in I : u, v \in X_i$.
3. $\forall i, j, k \in I$*: if j is on a path in T from i to k, then* $X_i \cap X_k \subseteq X_j$.

The width of a tree decomposition $(\{X_i : i \in I\}, T)$ is defined to be equal to $\max_{i \in I} |X_i| - 1$.

The treewidth of G is defined to be the minimum width over all tree decompositions of G and is denoted by $tw(G)$.

A path decomposition of $G = (V, E)$ is a tree decomposition (X, T) in which T is required to be a path. The pathwidth of G is defined to be the minimum width over all path decompositions of G and is denoted by $pw(G)$.

Recent research has shown that many NP-complete problems become polynomial or even linear time solvable, or belong to NC, when restricted to graphs with small treewidth (See [2, 3]). The decision problem of checking, given G and k, whether $tw(G) \leq k$ is known to be NP-complete. Hence the problem of determining the treewidth of an arbitrary graph is NP-hard.

Another interesting "width" parameter is the carving width. This notion is used by Seymour and Thomas in [23] in their algorithm to decide

whether there exists a routing tree with congestion less than k, when the underlying graph is planar. We present below the definition of carving width.

Let V be a finite set with $|V| \geq 2$. Two subsets $A, B \subseteq V$ cross if $A \cap B, A - B, B - A, V - (A \cup B)$ are all non–empty. A carving in V is a set $\mathcal{C}$ of subsets of V such that

1. $\emptyset, V \notin \mathcal{C}$
2. no two members of C cross
3. $\mathcal{C}$ is maximal subject to the two conditions above.

Let G be a graph. For $A \subset V(G)$, let $\delta(A)$ denote the set of all edges with exactly one end point in A. For each $e \in E(G)$, let $p(e) \geq 0$ be an integer. For $X \subseteq E(G)$, we denote $\sum_{e \in X} p(e)$ by $p(X)$. If $|V(G)| \geq 2$, we define the p–carving width of G to be the minimum over all carvings in $V(G)$, of the maximum over all $A \in \mathcal{C}$, of $p(\delta(A))$. The carving width of G is the p–carving width of G where $p(e) = 1$ for every edge e and is denoted by $cw(G)$. Seymour and Thomas show that computing $cw(G)$ for general graphs is NP-hard.

The d–dimensional hypercube, H_d, is the graph on 2^d vertices, which correspond to the 2^d d–vectors whose components are either 0 or 1, two of the vertices being adjacent when they differ in just one coordinate. Hypercubes are a well-studied class of graphs, which arise in the context of parallel computing, coding theory, algebraic graph theory and many other areas. They are popular because of their symmetry, small diameter and many interesting graph–theoretic properties.

1.1 Our Results

In this paper we develop lower bounds for treewidth, pathwidth and carving width of a graph in terms of isoperimetric inequalities. As applications of these lower bounds we have the following results.

Result 1 The length of the shortest cycle in a graph G is called the girth of G. We derive a lower bound for the treewidth of G in terms of its girth and average degree. In particular, we show that if G has girth g and average degree d, then $tw(G) \geq \max\left(\left\lceil \frac{d}{2} \right\rceil, \ \frac{1}{4e(g+1)}(d-1)^{\lfloor (g-1)/2 \rfloor} - 2\right)$, where e is the Napier number. In view of a well–known conjecture regarding the existence of graphs with girth g, minimum degree δ and having at most $c(\delta - 1)^{\lfloor (g-1)/2 \rfloor}$ vertices (for some constant c), this lower bound seems to be almost tight (but for a multiplicative factor of $g + 1$). (See section 2 for details.)

Result 2 We study the parameters treewidth, pathwidth and carving width in the context of d–dimensional hyper cubes. The question of estimating the treewidth of hyper cube was proposed by Chlebikova [10] who states that $2^{\lfloor \frac{d}{2} \rfloor} \leq tw(H_d) \leq \binom{d}{\lfloor \frac{d}{2} \rfloor} + \binom{d}{\lceil \frac{d}{2} \rceil}$. In [9], Chandran and Subramanian show that $\text{tw}(H_d) \geq \left\lfloor \frac{3.2^d}{2(d+4)} \right\rfloor - 1$. In this paper we obtain improved bounds for the treewidth of H_d and determine the exact values for the pathwidth and carving width of H_d.

1. $pw(H_d) = \sum_{m=0}^{d-1} \binom{m}{\frac{m}{2}}$.
2. $c_1 \frac{2^d}{\sqrt{d}} \geq tw(H_d) \geq c_2 \frac{2^d}{\sqrt{d}}$, for some constants c_1, c_2, where $c_1 < 1.2$ and $c_2 > 0.48$, for large d.
3. $cw(H_d) = 2^{d-1}$.

Result 3 In fact, we study these parameters in the context of a more general class of graphs, namely the Hamming graphs. The *Hamming Graph* K_q^d is the graph on q^d vertices, which correspond to the q^d d–vectors with coordinates coming from a set of size q, two of the vertices being adjacent if they differ in just one coordinate. (Clearly, the d–dimensional hypercubes are a special case of the Hamming graphs, K_q^d, namely, when $q = 2$).

1. $pw(K_q^d) \geq c_1 \frac{q^d}{\sqrt{d}}$ for some constant $c_1 > 0$, where $c_1 \approx \sqrt{\frac{2}{\pi}} e^{-2}$ for large d.
2. $pw(K_q^d) \leq c_2 \frac{q^d}{\sqrt{d}}$ for some constant c_2, where $c_2 < 1.2$ for large d.

2 A lower bound for treewidth and an application

Definition 2. *Let $G = (V, E)$ be a simple connected graph on n vertices with no self loops, and s be a number such that $1 \leq s \leq n$. We define $N_{min}(s) = \min |N(X)|$ over all (non empty) X with $\frac{s}{2} \leq |X| \leq s$.*

It can be shown that $tw(G) \geq N_{min}(s) - 1$, for any s, $1 \leq s \leq |V(G)|$. The details of the proof of this result is is available in the full paper [8]. Noting that $N_{min}(s) = \min_{\frac{s}{2} \leq x \leq s} b_v(x, G)$, we can state the following theorem.

Theorem 1. *Let $G = (V, E)$ be a graph and let s be an integer with $1 \leq s \leq n$. Then $tw(G) \geq \min_{\frac{s}{2} \leq x \leq s} b_v(x, G) - 1$.*

We make use of this lower bound for treewidth to derive the following result. We omit the proof. (It is available in the full paper [8]).

Theorem 2. *Let G be a graph with girth at least g, average degree at least d. Then*

$$tw(G) \geq \max\left(\left\lceil \frac{d}{2} \right\rceil, \quad \frac{1}{4e(g+1)}(d-1)^{\lfloor (g-1)/2 \rfloor} - 2\right)$$

where e is the Napier number.

Perhaps, a natural approach to the question of getting a lower bound for the treewidth of G would be to look for the largest r such that G has a K_r minor (K_r stands for the complete graph on r vertices), since the treewidth of a graph is at least the treewidth of any of its minors and $tw(K_r) = r - 1$. (H is called a minor of G iff H can be obtained from a subgraph of G by a sequence of edge contractions). This problem has received some attention in the literature. It can be inferred from a result of Thomassen [25] in conjunction with a result [1] in [17, 24] that if $girth(G)$ is at least $cr\sqrt{\log r}$ and the minimum degree of G is at least 3 then G has a K_r minor. This result was improved recently by Diestel and Rempel [12] who proved that if $girth(G) \geq 6 \log r + 3 \log \log r + c$, and $\delta(G) \geq 3$, then G has a K_r minor. One can infer from the above result that if $\delta(G) \geq 3$, then $tw(G) \geq r' - 1 \geq 2^{g/6} - 1$, where r' is the largest integer such that G has a $K_{r'}$ minor. Very recently Kühn and Osthus [18] have improved his result, and have shown using probabilistic methods that if $girth(G) \geq g$ for some odd g, then for some $c > 0$, G has a K_r minor for some $r \geq \frac{c.(\delta)^{\frac{g+1}{4}}}{\sqrt{\log \delta}}$. Clearly, the lower bound we provide for treewidth in terms of girth and minimum degree is much better than (in fact about the square of) what is derivable from the minor theoretic result of Kühn and Osthus. Moreover, our lower bound is expressed in terms of average degree rather than the minimum degree and thus is stronger.

More fundamentally, Kühn and Osthus argue that it is unlikely that their lower bound for the clique minor size may be improved significantly. Thus, extending their point further, it is unlikely that an approach based on clique minors may give a lower bound for the treewidth (in terms of girth and minimum degree or average degree) comparable to ours.

For a brief exposition of the early developments on the existence of dense minors in graphs of large girth, see Chapter 8 of [11]. For an introductory account of the role of treewidth in minor theory, see Chapter 12 of [11].

[1] The result of [17, 24] states that if the average degree of a graph G is at least $cr\sqrt{\log r}$ then G has a K_r minor.

Remark on the tightness of the above lower bound: The following is a well known conjecture on the existence of high girth graphs. (See for example, [5], page 164).

Conjecture: *There exists a constant c such that for all integers $g, \delta \geq 3$, there is a graph $G(g, \delta)$ of minimum degree at least δ and girth at least g whose order (number of vertices) is at most $c(\delta - 1)^{\lfloor \frac{g-1}{2} \rfloor}$.*

In view of this conjecture, we see that the lower bound given in Theorem 2 is very close to what is best possible (but for a multiplicative factor of $g + 1$) since the treewidth of $G(g, \delta)$ can be at most $c(\delta - 1)^{\lfloor \frac{g-1}{2} \rfloor}$, the total number of vertices, whereas the lower bound proven by us is at least $\frac{1}{4e(g+1)}(\delta - 1)^{\lfloor (g-1)/2 \rfloor} - 2$ since the average degree $d \geq \delta$. It may be noted that, by using $G(g, \delta)$ as components it is easy to construct graphs with minimum degree at least δ, girth at least g, and having an arbitrarily large number of vertices but its treewidth (or pathwidth) still at most $c(\delta - 1)^{\lfloor \frac{g-1}{2} \rfloor}$.

3 A lower bound for pathwidth and carving width

Since the $tw(G) \leq pw(G)$, we can, of course, use Theorem 1 to provide a lower bound for the pathwidth of a graph also. But we notice that in the case of pathwidth one can state a much stronger result. Below we develop the necessary concepts.

A graph $G = (V, E)$ is defined to be an interval graph iff its vertices $V = \{v_1, v_2, \cdots, v_n\}$ can be put in one to one correspondence with a set of intervals $\{I_1, I_2, \cdots, I_n\}$ on the real line in such a way that $\{v_i, v_j\} \in E$ if and only if the corresponding intervals I_i and I_j have a non empty intersection. Without loss of generality one can assume that all the intervals are closed intervals, see for example [14], page 13. The reader can easily convince himself that given any graph G there exist super graphs G' of G such that G' is an interval graph. For example, the complete graph on n nodes is an interval graph. The clique number $\omega(G)$ is defined to be the number of vertices in a maximum sized clique in G.

Definition 3. *A graph G has interval–width, $IW(G) = k$ iff k is the smallest non–negative integer such that G is a subgraph of some interval graph H, with $\omega(H) = k + 1$.*

The following characterization of pathwidth in terms of interval graphs is well–known. The omitted proofs (from this section onwards) are available in the full paper [7].

Lemma 1. *Let G be a graph. Then $pw(G) = IW(G)$.*

Lemma 2. *Let $G = (V, E)$ be a connected interval graph and $1 \leq s \leq n$. Then $\omega(G) \geq b_v(s, G) + 1$.*

Theorem 3. *Let $G = (V, E)$ be any graph on n nodes, and let $1 \leq s \leq n$. Then $pw(G) \geq b_v(s, G)$.*

Proof. Clearly in any interval super graph G' of G (on the same number of vertices) $b_v(s, G') \geq b_v(s, G)$. The theorem follows from Lemma 1 and Lemma 2. ■

A lower bound similar to that of Theorem 1 can be proved for the carving width of a graph G in terms of its edge isoperimetry. The following lemma is from [23].

Lemma 3 ([23]). *Let V be a finite set with $|V| \geq 2$, let T be a tree in which every vertex has valency 1 or 3, and let τ be a bijection from V onto the set of leaves of T. For each edge e of T let $T_1(e)$ and $T_2(e)$ be the two components of $T - e$; and let*

$$\mathcal{C} = \{\{v \in V : \tau(v) \in V(T_i(e))\} : e \in E(T), i = 1, 2\}$$

Then $\mathcal{C}$ is a carving in V. Conversely, every carving in V arises from some tree T and bijection τ in this way.

Lemma 4. *Let T be a tree with each of its nodes having valency either 3 or 1. For any edge e, let $T_1(e)$ and $T_2(e)$ be the components of $T \backslash e$. For any $1 \leq x \leq n, \exists$ an edge e_x such that the number of leaves of one of the trees $T_1(e_x)$ and $T_2(e_x)$ lies between $\frac{x}{2}$ and x.*

Theorem 4. *Let $G = (V, E)$ be a graph on n nodes and let $1 \leq x \leq n$. Then $cw(G) \geq \min_{\frac{x}{2} \leq s \leq x} b_e(s, G)$.*

4 Applications to hypercubes and hamming graphs

4.1 Hypercube

A prime example of a graph of combinatorial interest is the hypercube H_d. The reader may be reminded that H_d can be considered as a graph on the power set $\mathcal{P}(X)$ of a d-point set X in which a vertex corresponding to a set x is adjacent to a vertex corresponding to a set y if and only if $|x \Delta y| = 1$. (Here $x \Delta y$ stands for the symmetric difference of the sets x and y.)

Before we deal with the pathwidth, treewidth and carving width of the hypercube, we introduce another well-known graph theoretic parameter, namely the bandwidth. Let $G = (V, E)$ be a graph. Let a bijection $\phi : V \to \{1, \cdots, n\}$ be called an ordering of the vertices of G. Then for any edge $e = \{u, v\} \in E$, let $\Delta(e, \phi) = |\phi(u) - \phi(v)|$. The bandwidth of G is defined as the minimum over all possible orderings ϕ of $V(G)$, the maximum value of $\Delta(e, \phi)$ over all edges $e \in E$. The following is easy to prove.

$$\text{bandwidth}(G) \geq \max_{1 \leq s \leq n} b_v(s, G) \tag{3}$$

Also, it is not difficult to see from the definition of pathwidth and bandwidth that $bandwidth(G) \geq pw(G)$. Note that this inequality can be strict and the value of $|bandwidth(G) - pw(G)|$ can be arbitrarily large. Below we will show that in the case of H_d, pathwidth=bandwidth.

In [15], Harper addressed the question of finding the bandwidth of H_d. To minimize the value of $\max_{e \in E} \Delta(e, \phi)$ he proposed an optimal ordering ϕ' of the vertices of H_d. He indeed showed that $\max_e \Delta(e, \phi') = max_s b_v(s, H_d)$. Harper's result on the bandwidth of H_d can be summarised as below.

Lemma 5. *$bandwidth(H_d) = \max_s b_v(s, H_d) = \sum_{m=0}^{d-1} \binom{m}{\frac{m}{2}}$*

The last equality can be proved by induction and is mentioned in the last page of Harper's article [15]. Putting together all the pieces, we get our result on the pathwidth of H_d.

Theorem 5. *$pw(H_d) = bandwidth(H_d) = \max_s b_v(s, H_d) = \sum_{m=0}^{d-1} \binom{m}{\frac{m}{2}}$*

Proof. By Theorem 3, we know that $\text{pw}(H_d) \geq \max_s b_v(s, H_d)$. It follows by Lemma 5, that $\text{pw}(H_d) \geq \text{bandwidth}(H_d)$. But we know that bandwidth of any graph is at least as much as its pathwidth. The required result follows. ■

The following result was first proved by Harper [15]. Simpler proofs were later given by Katona [16], Frankl and Füredi [13] etc. See [4], Chapter 16 for an exposition.

Lemma 6. *Let s be an integer such that $s = \sum_{i=0}^{r-1} \binom{d}{i} + m$, where $0 < m \leq \binom{d}{r}$ $(1 \leq r \leq d.)$ Then, $b_v(s, H_d) = \binom{d}{r} - m + \partial_u(m)$ where $\partial_u(m)$ is the minimum possible cardinality of the upper shadow of a set of m subsets in X^r (see [4], chapter 5 for details).*

By setting $m = \binom{d}{r}$ in the above lemma, we have the following simple corollary.

Corollary 1. $b_v(\sum_{i=0}^{r}\binom{d}{r}, H_d) = \binom{d}{r+1}$

Now, making use of Lemma 6, Corollary 1 and Theorem 1 and using sterling's approximation for factorials, we can present our bounds for the treewidth of H_d.

Theorem 6. *$c_1\frac{2^d}{\sqrt{d}} \geq tw(H_d) \geq c_2\frac{2^d}{\sqrt{d}}$, for some constants c_1, c_2, where $c_1 < 1.2$ and $c_2 > 0.48$, for large d.*

Similarly, using Theorem 4 we can infer the following result.

Theorem 7. *$cw(H_d) = 2^{d-1}$.*

Proof. It follows from Bollobás [4] and Bollobás and Leader [6] that $\min_{2^{d-2}\leq s\leq 2^{d-1}} b_e(s, H_d) \geq 2^{d-1}$. So, by Theorem 4, we get that $cw(H_d) \geq 2^{d-1}$. In fact we can construct a carving $\mathcal{C}$ on H_d with width $= 2^{d-1}$, thus proving the theorem. (See the full paper [7] for details.) ■

4.2 Hamming Graphs

Another graph of combinatorial interest is K_q^d, the product of d copies of a complete graph of order q. Equivalently, the vertex set of K_q^d is the set of all strings of size d on the alphabet $\{1, 2, ..., q\}$ and two strings are adjacent if they differ in exactly one place.

Theorem 8. *$pw(K_q^d) \geq c_1\frac{q^d}{\sqrt{d}}$ for some constant $c_1 > 0$, where $c_1 \approx \sqrt{\frac{2}{\pi}}e^{-2}$ for large d.*

Theorem 9. *$pw(K_q^d) \leq c_2\frac{q^d}{\sqrt{d}}$ for some constant c_2, where $c_2 < 1.2$ for large d.*

Combining Theorems 8 and 9, we see that the $pw(K_q^d) = \Theta(\frac{q^d}{\sqrt{d}})$.

References

1. N. Alon and V. D. Millman, λ_1, *isoperimetric inequalities for graphs and super concentrators*, Journal of Combinatorial Theory, Series. B, 38 (1985), pp. 73–88.
2. S. Arnborg and A. Proskurowski, *Linear time algorithms for NP–hard problems on graphs embedded in k–trees*, Discrete Applied Mathematics, 23 (1989), pp. 11–24.

3. H. L. Bodlaender, *A tourist guide through treewidth*, Acta Cybernetica, 11 (1993), pp. 1–21.
4. B. Bollabás, *Combinatorics*, Cambridge University Press, 1986.
5. B. Bollobás, *Extremal Graph Theory*, Academic Press, 1978.
6. B. Bollobás and I. Leader, *Edge-isoperimetric inequalities in the grid*, Combinatorica, 11 (1991), pp. 299–314.
7. L. S. Chandran and T. Kavitha, *Lower bounds for width parameters of graphs using isoperimetric inequalities*. Manuscript, 2003.
8. L. S. Chandran and C. R. Subramanian, *Girth and treewidth*, Tech. Rep. MPI-I-2003-NWG2-01, Max-Planck-Institut für Informatik, Saarbrücken, Germany, 2003.
9. ———, *A spectral lower bound for the treewidth of a graph and its consequences*. To appear in Information Processing Letters, 2003.
10. J. Chlebikova, *On the tree-width of a graph*, Acta Mathematica Universitatis Comenianae, 61 (1992), pp. 225–236.
11. R. Diestel, *Graph Theory*, vol. 173, Springer Verlag, New York, 2 ed., 2000.
12. R. Diestel and C. Rempel, *Dense minors in graphs of large girth*. To Appear in Combinatorica, 2003.
13. P. Frankl and Z. Füredi, *A short proof for a theorem of Harper about Hamming spheres*, Discrete Math., 34 (1981), pp. 311–313.
14. M. C. Golumbic, *Algorithmic Graph Theory And Perfect Graphs*, Academic Press, New York, 1980.
15. L. Harper, *Optimal numberings and isoperimetric problems on graphs*, Journal of Combinatorial Theory, 1 (1966), pp. 385–393.
16. G. O. H. Katona, *The hamming-sphere has minimum boundary*, Studia Sci. Math. Hungar., 10 (1975), pp. 131–140.
17. A. V. Kostochka, *Lower bound of the hadwiger number of graphs by their average degree*, Combinatorica, 4 (1984), pp. 307–316.
18. D. Kühn and D. Osthus, *Minors in graphs of large girth*. To appear in Random Structures and Algorithms, 2003.
19. I. Leader, *Discrete isoperimetric inequalities*, Proc. Symp. Appl. Math., 44 (1991), pp. 57–80.
20. N. Robertson and P. D. Seymour, *Graph minors I. excluding a forest*, Journal of Combinatorial Theory, Ser. B, 35 (1983), pp. 39–61.
21. ———, *Graph minors II: algorithmic aspects of tree-width*, Journal of Algorithms, 7 (1986), pp. 309–322.
22. G. Schechtman, *Lévy type inequality for a class of metric spaces*, in Martingale Theory in Harmonic Analysis and Banach Spaces, Lecture Notes in Mathematics, vol. 939, 1982, pp. 211–215.
23. P. Seymour and R. Thomas, *Call routing and the ratcatcher*, Combinatorica, 14 (1994), pp. 217–241.
24. A. G. Thomason, *An extremal function for contractions of graphs*, Math. Proc. Camb. Phil. Soc., 95 (1984), pp. 261–265.
25. C. Thomassen, *Girth in graphs*, Journal of Combinatorial Theory, Ser. B, 35 (1983), pp. 129–141.